



Philippine Christian University

Sampaloc I, Dasmariñas, Cavite

<https://dasma.pcu.edu.ph>

Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server) ss
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running) alxa
2. Go to **Settings** → **Network** sud
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM

2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SHH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 83386sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:                0.0
Usage of /:                  5.1% of 48.91GB
Memory usage:               5%
Swap usage:                 0%
Processes:                  139
Users logged in:            1
IPv4 address for enp0s3:    192.168.1.14
IPv6 address for enp0s3:    2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

Name: Ravelo, Alex G.

Student ID: 2024300917

Course/Year/Section: BSCS 3A

Date Submitted: 03/09/2026

Instructor: Johneros P. Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☒ ~~Create and configure a Virtual Machine with Ubuntu Server~~
- ☒ ~~Configure Bridged Network Adapter for network access~~
- ☒ ~~Determine and verify the server's IP address~~
- ☒ ~~Install OpenSSH Server and enable the service~~
- ☒ ~~Establish successful SSH connection from host machine~~
- ☒ ~~Execute verification commands to confirm remote access~~

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu (64 bit)
Virtualization Software (VirtualBox/VMware)	VirtualBox
Ubuntu Server Version	Ubuntu-26.04
VM Name	Ravelo-VM
RAM Allocated	8 GB
Storage Allocated	100 GB
Network Mode	NAT
SSH Client Used	OpenBSD Secure Shell

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	8 MB
Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

Server IP Address: 192.254.83.68

Network Interface: Physical

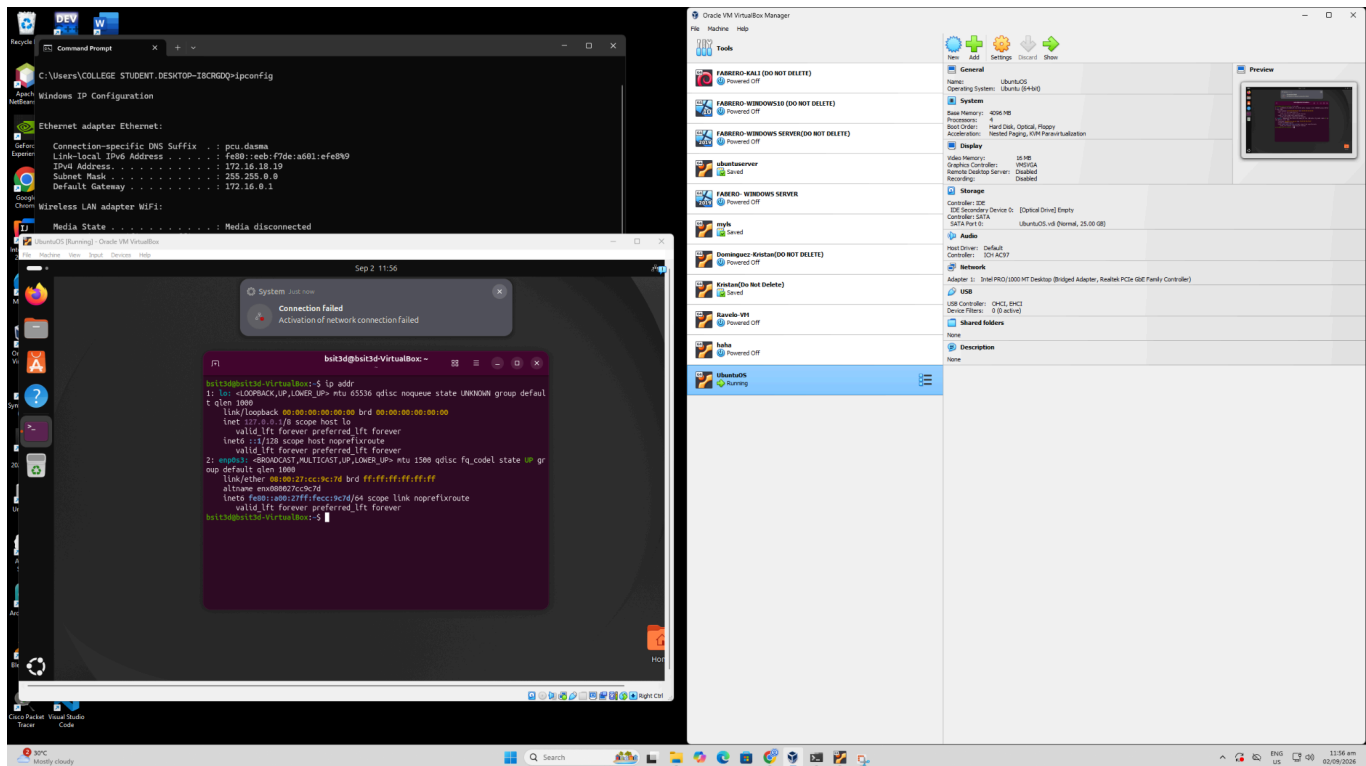
6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.254.83.68
Username	alexa
SSH Port	16

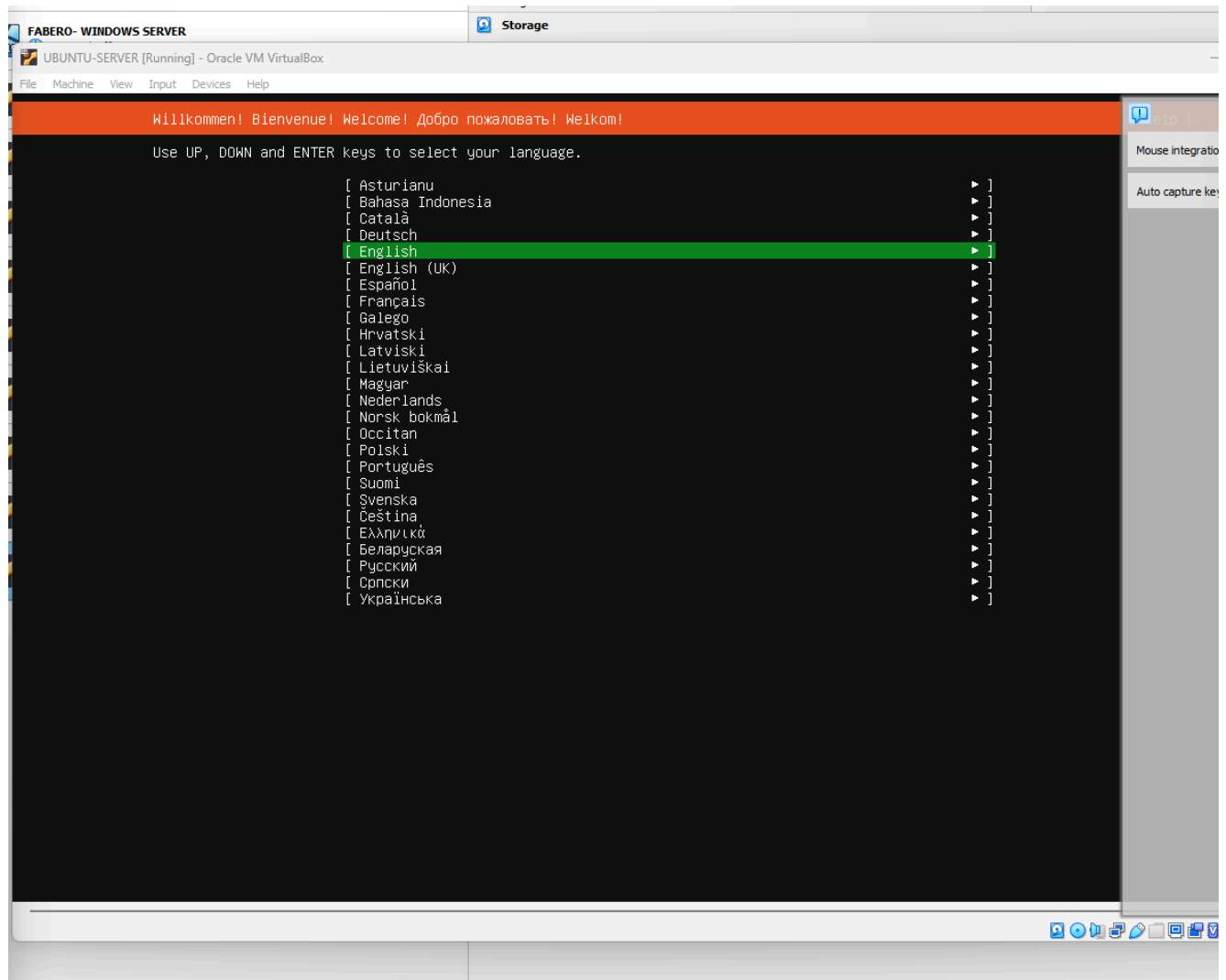
Open VirtualBox

- Open virtualbox and create new then select ubuntu live server.



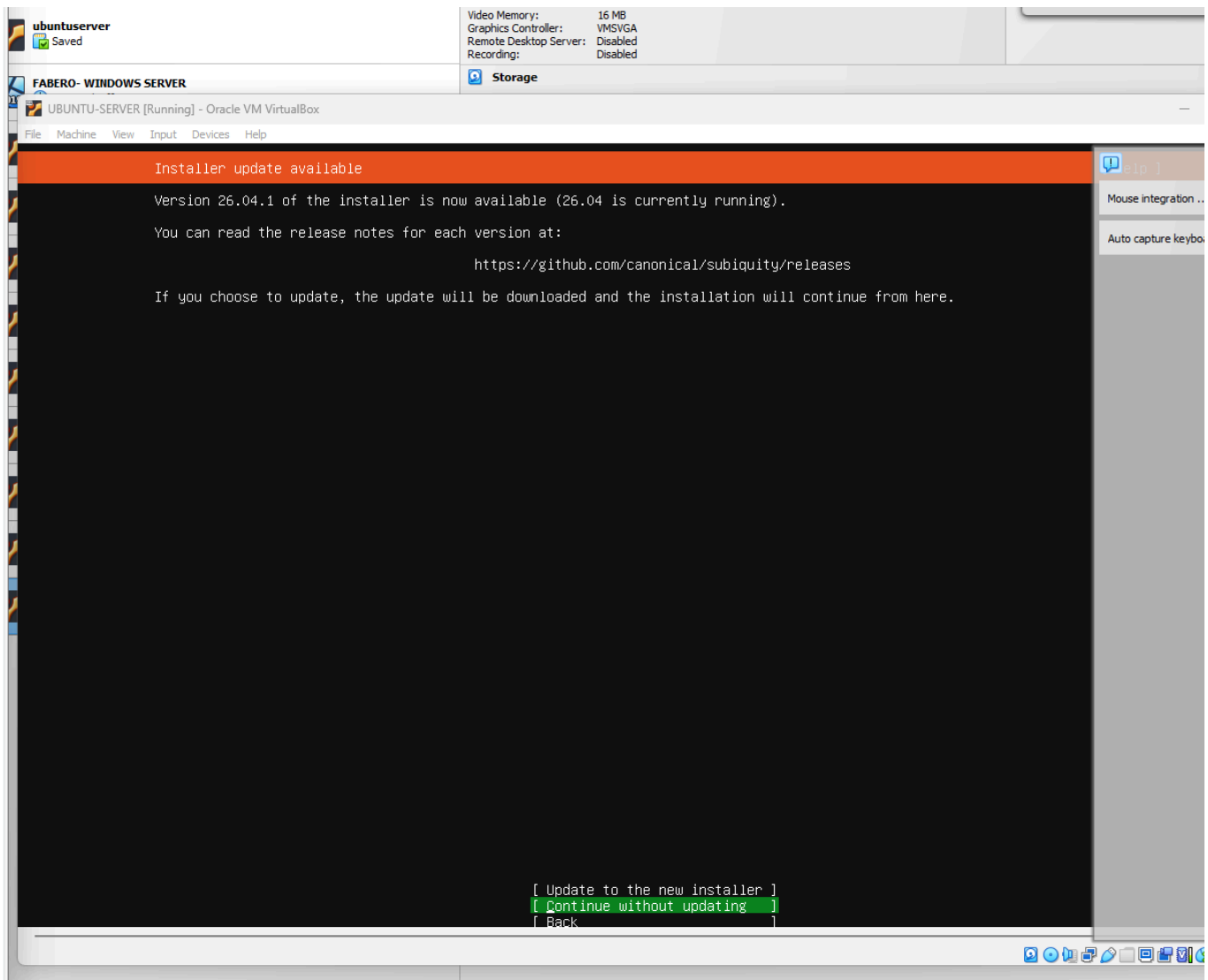
SELECT

- After creating the server continue on working or configuring the running server.



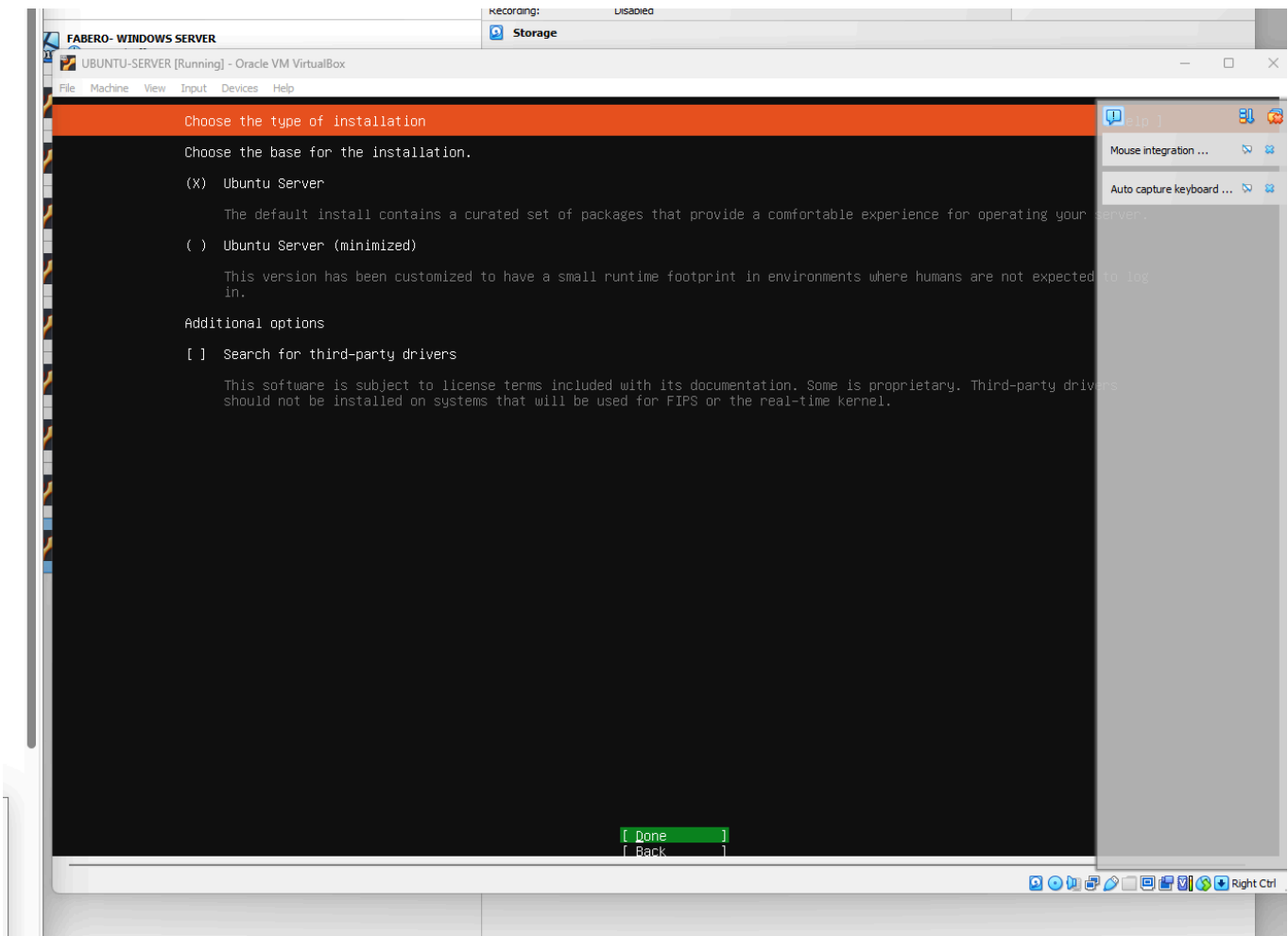
Next

- Select Continue without updating.



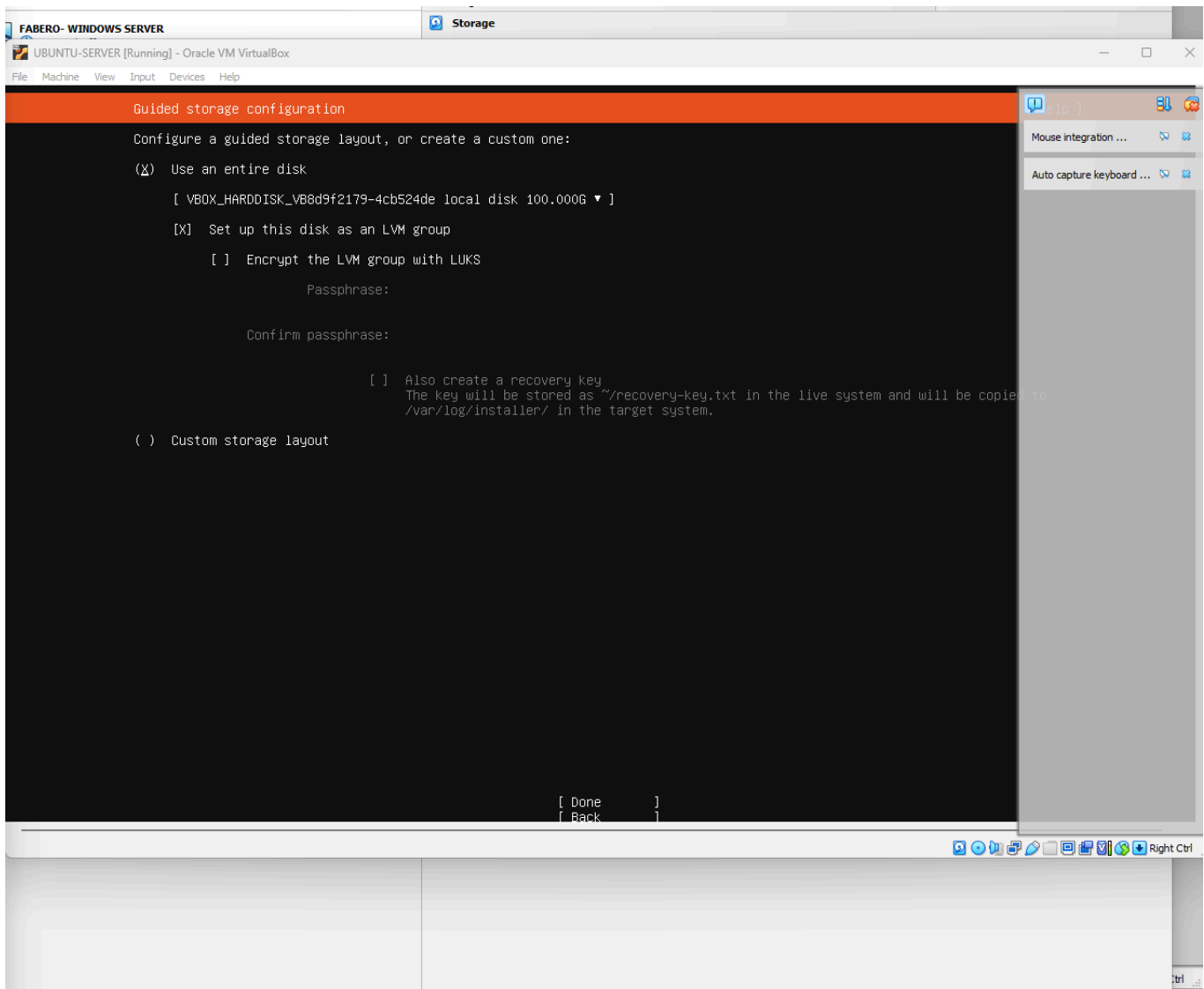
Next

- Proceed to the next step don't click anything but done.



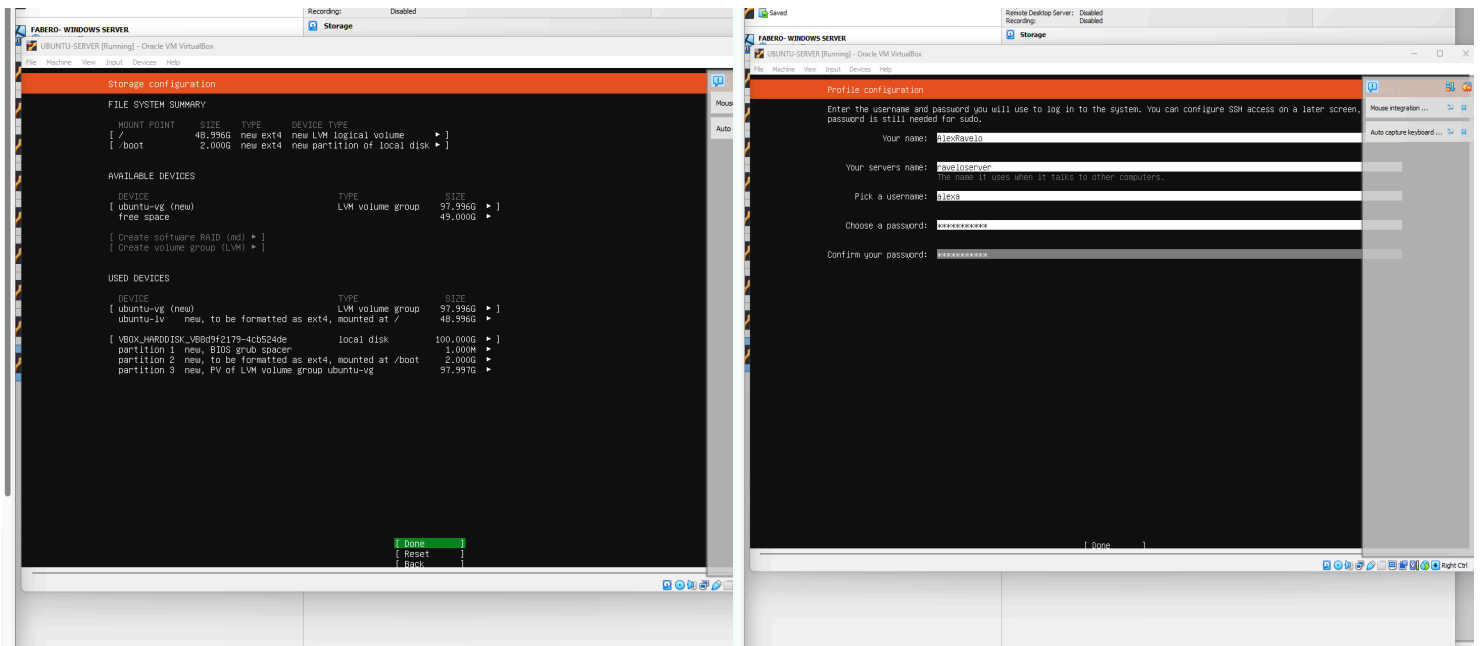
Proceed

- Proceed to click done and don't (x) anything.



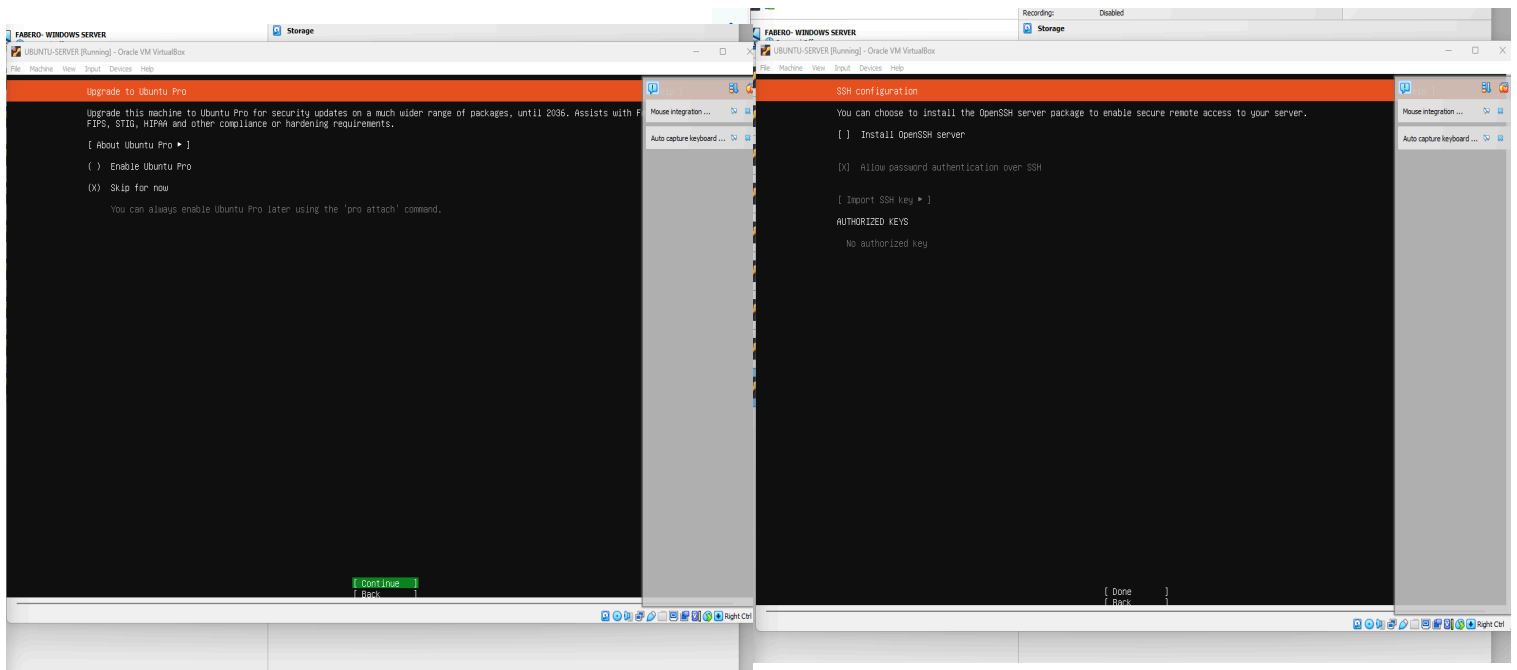
Next

- when storage configuration pops up proceed to done.
- Profile configuration fill the needed information.



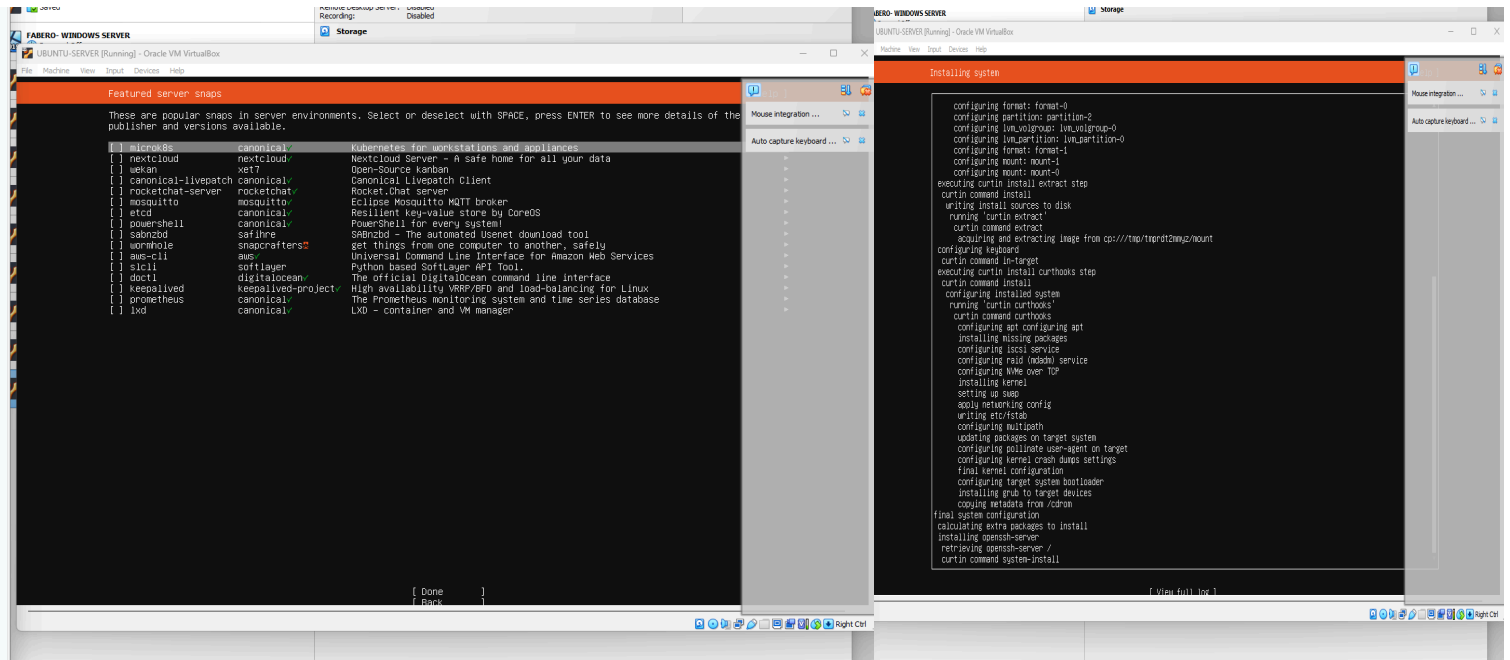
Upgrade/SSH

- When upgrade configuration pops up proceed to done make sure its "skip for now"
- SSH configuration make sure to select (x) to install OPENSSSH then proceed to done.



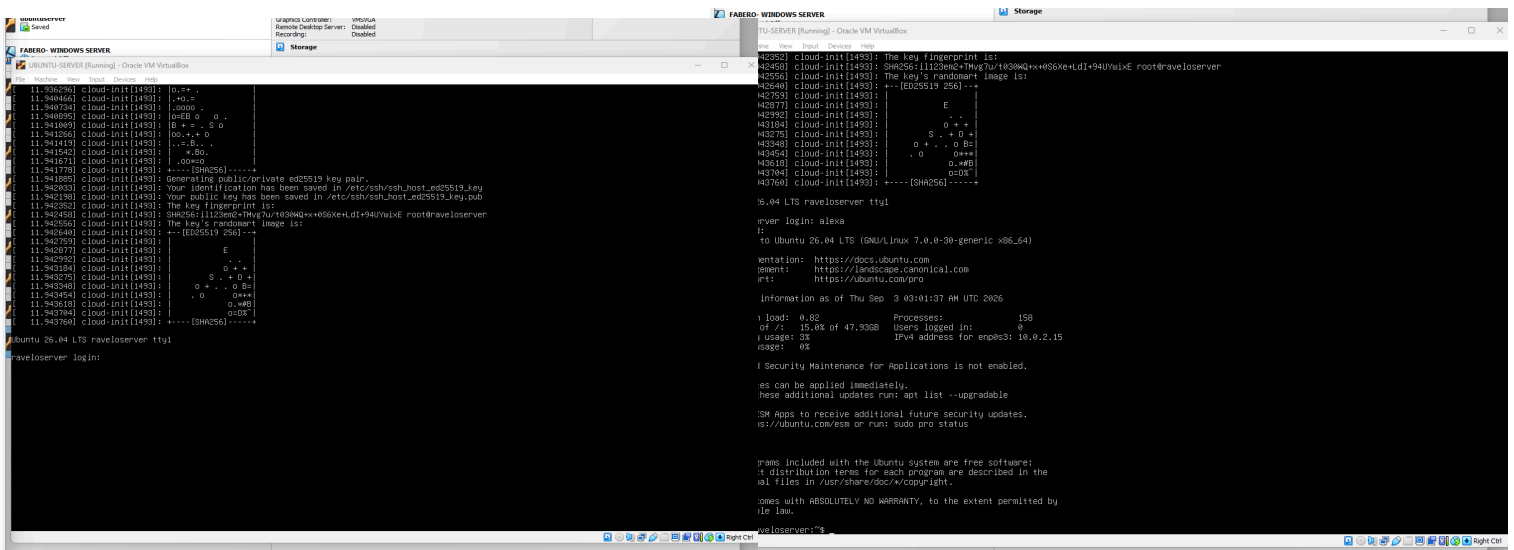
Featured/Installing

- Double check on the information on the features and if its all good proceed to done.
- Once you selected done the installing system will pop up do not click anything and wait for a reboot to show up in the bottom below the [View Full Log]. When it shows up click reboot then proceed to the next step.



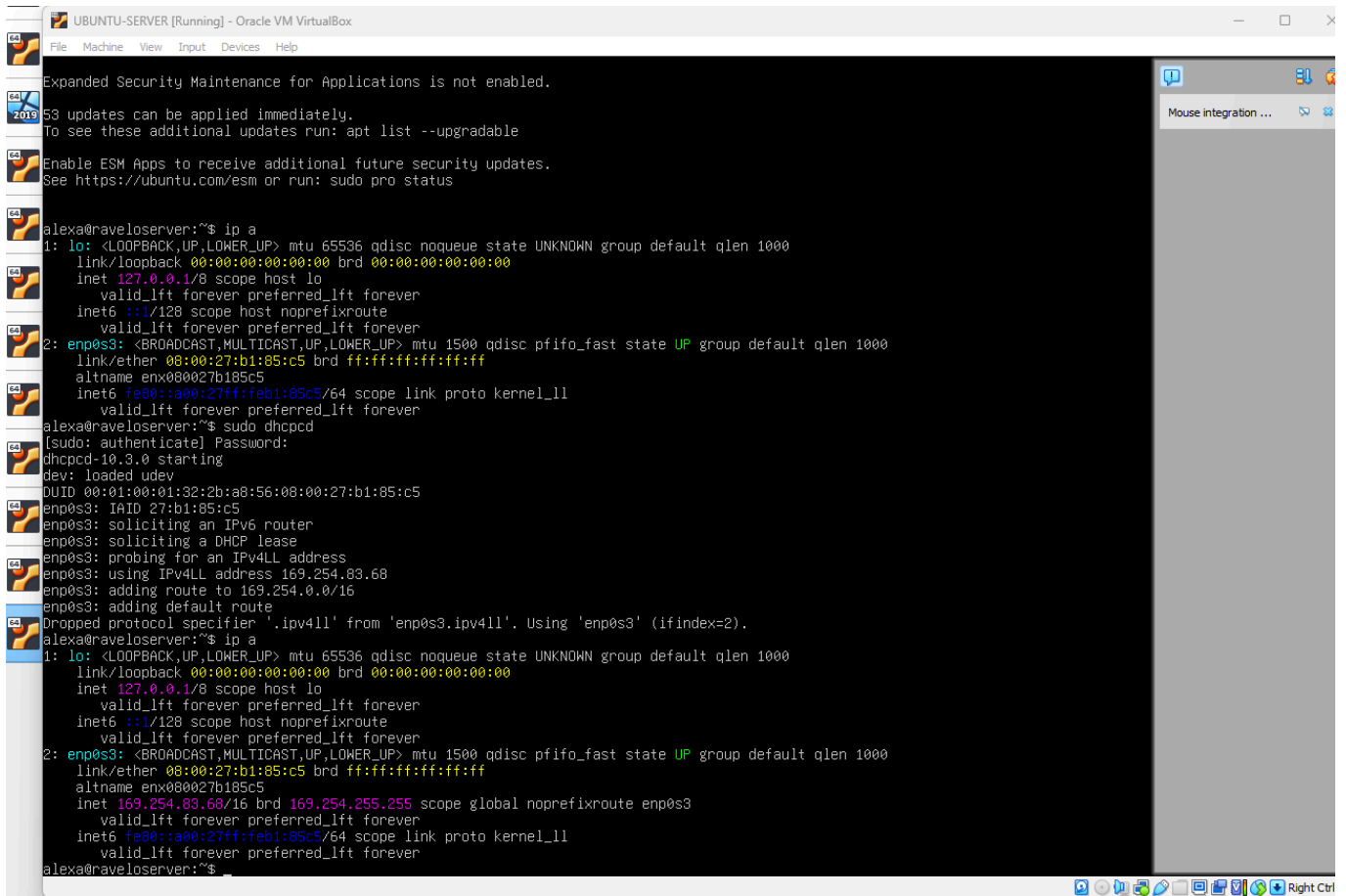
Login/Command

- Once rebooted you can now login and input password.
- After inputting the password you can now input the instruction or command to show the needed information. (can be seen below)

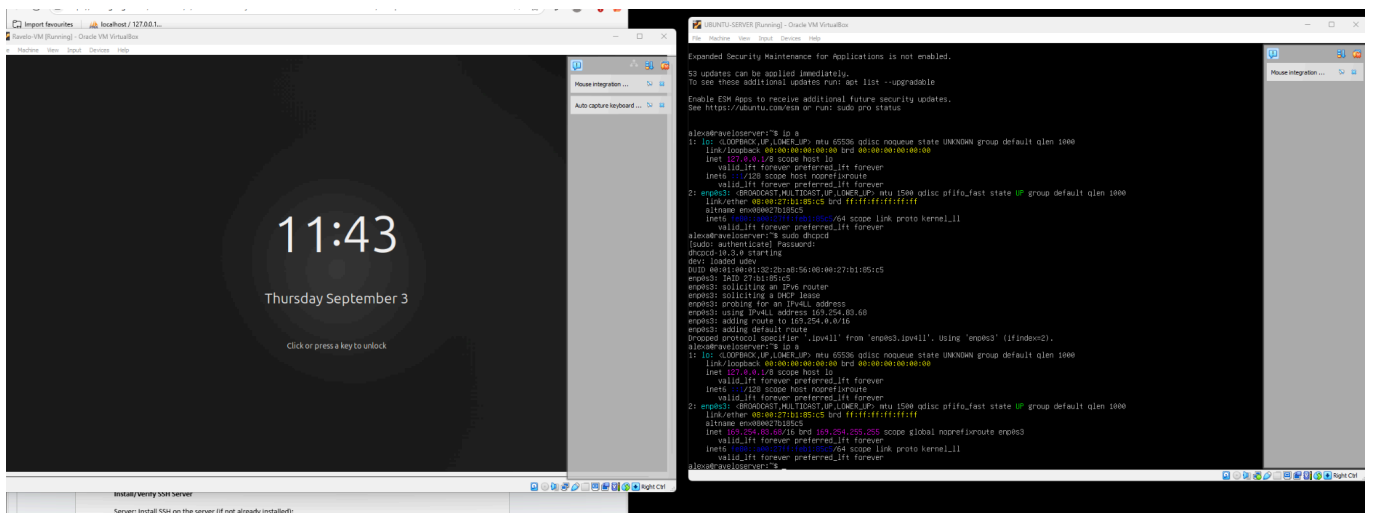


IP ADD

- Check IP address and then proceed to open your ubuntu and go to terminal and input your ip address along with your username.



```
alex@raveloserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:b1:85:c5 brd ff:ff:ff:ff:ff:ff
    altname enx080027b185c5
    inet6 fe80::a00:27ff:feb1:85c5/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
alex@raveloserver:~$ sudo dhcpcd
[sudo: authenticate] Password:
dhcpcd-10.3.0 starting
dev: loaded udev
DUID 00:01:00:01:32:b8:56:08:00:27:b1:85:c5
enp0s3: IAID 27:b1:85:c5
enp0s3: soliciting an IPv6 router
enp0s3: soliciting a DHCP lease
enp0s3: probing for an IPv4LL address
enp0s3: using IPv4LL address 169.254.83.68
enp0s3: adding route to 169.254.0.0/16
enp0s3: adding default route
Dropped protocol specifier '.ipv4ll' from 'enp0s3.ipv4ll'. Using 'enp0s3' (ifindex=2).
alex@raveloserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:b1:85:c5 brd ff:ff:ff:ff:ff:ff
    altname enx080027b185c5
    inet 169.254.83.68/16 brd 169.254.255.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feb1:85c5/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
alex@raveloserver:~$
```



```
alex-g-ravelo@alex-g-ravelo-VirtualBox:~$ ssh alexa@169.254.83.68
alex-g-ravelo@alex-g-ravelo-VirtualBox:~$ ssh alexa@169.254.83.68
alex@169.254.83.68's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 03:55:14 AM UTC 2026

System load: 0.02          Memory usage: 3%    Processes:   160
Usage of /: 15.1% of 47.93GB Swap usage: 0%      Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

alex@raveloserver:~$
```